
MTH302 MIDTERM EXAMINATION
Spring 2011
(VISIT VU RANK FOR MORE)

Question No: 1 (Marks: 1) - Please choose one

A, B and C play cricket. A's runs are to B's runs and B's runs are to C's as 3:2. They get altogether 342 runs.

How many runs did A make?

- ▶ **162**
- ▶ 108
- ▶ 72
- ▶ 112

Question No: 2 (Marks: 1) - Please choose one

_____ The price of a P4 computer was \$12000 last year. What is the price of the computer for this year if the price is decreased by 10%?

- ▶ \$12000
- ▶ \$12500
- ▶ **\$10800**
- ▶ \$9000

Question No: 3 (Marks: 1) - Please choose one

_____ Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true?

- ▶ The mean is 103.
- ▶ **The mean is 104.**
- ▶ The median is 100.
- ▶ The median is 106.

Question No: 4 (Marks: 1) - Please choose one

Percentage is a ratio of given number with standard number:

- ▶ 100
- ▶ Same number
- ▶ 1000
- ▶ 10

Question No: 5 (Marks: 1) - Please choose one

The Virtual University of Pakistan is a growing distant learning university of Pakistan. The number of students in the university is increasing each year. The number of registered students in a course related to business math last year was 7113 and it is increased up to 9242 this year. What is the percentage increase in the number of students?

- ▶ 23%
- ▶ 30%
- ▶ 18%
- ▶ 42%

Question No: 6 (Marks: 1) - Please choose one

If A and B are any two matrices of order $m \times n$ and $p \times q$ respectively and $m > n$ and $p < q$. What should be the condition on m, n, p, q for the product AB to hold?

- ▶ $n > p$
- ▶ $m \leq q$
- ▶ $q = p$
- ▶ $n = p$

Question No: 7 (Marks: 1) - Please choose one

The price at which investors buy or sell a share of stock at a given time is referred as

- ▶ Face value
- ▶ Market value
- ▶ Accumulated value
- ▶ Earning value

Question No: 8 (Marks: 1) - Please choose one

Gross salary is also called Net salary

- ▶ True
- ▶ False

Question No: 9 (Marks: 1) - Please choose one

Can

we subtract a positive integer k from the identity matrix?

- ▶ Yes
- ▶ No

Question No: 10 (Marks: 1) - Please choose one

Net

price =

- ▶ List price + trade discount
- ▶ List price - trade discount
- ▶ List price / trade discount
- ▶ List price * trade discount

Question No: 11 (Marks: 1) - Please choose one

Division. Symbol: / is a type of

- ▶ Comparison operators
- ▶ Concatenation operator
- ▶ Reference operators
- ▶ Arithmetic operators

Question No: 12 (Marks: 1) - Please choose one

Given : List Price = 5500Rs , Discount = 850Rs, then Net Cost Price will be

▶ 6350 Rs

6350 Rs

▶ 5508 Rs

5508 Rs

- ▶ 5585 Rs
- ▶ 4650 Rs

Question No: 13 (Marks: 1) - Please choose one

_____ To
convert a number in percent (%) from a fraction, we do which of the following operation:

- ▶ add by 100
- ▶ subtract by 100
- ▶ multiply by 100
- ▶ divide by 100

Question No: 14 (Marks: 1) - Please choose one

_____ If
 A is the matrix of dimension 2×3 and I is the identity matrix of dimension 3×3 . Then which of the following is true

- ▶ $AI = I$
- ▶ $AI = A$
- ▶ $A + I = A$
- ▶ $A + I = I$

Question No: 15 (Marks: 1) - Please choose one

_____ If
ratio of boys and girls in a class is 7: 5, which of the following can't be the total number of the students in the class?

- ▶ 36
- ▶ 50
- ▶ 60
- ▶ 120

Question No: 16 (Marks: 1) - Please choose one

_____ *SLN*
returns the straight-line depreciation of an asset for

- ▶ Zero period

► One period

► Two periods

► Three periods

Question No: 17 (Marks: 1) - Please choose one

Assume you will work for the next 20 years and at the end of each year you will deposit \$20,000 into a savings account. At the end of that 20 year period you will start withdrawing your retirement living expenses from that account at the beginning of every year. This financial plan involves a _____ problem while you are working and a _____ problem while you are in retirement.

► a present value of a dollar, future value of a dollar

► a future value of an annuity due, a present value of an annuity due

► a future value of an annuity due, a present value of an ordinary annuity

► a future value of an ordinary annuity, a present value of an annuity due

► a future value of an ordinary annuity, a present value of an ordinary annuity

Question No: 18 (Marks: 1) - Please choose one

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 & 3 \\ 2 & 1 & 1 \end{bmatrix}$$

By multiplying the above two matrices we get the following result

$$C = \begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}$$

►

$$C = \begin{bmatrix} 9 \\ 7 \\ 17 \end{bmatrix}$$

►

$$C = \begin{bmatrix} 9 \\ 12 \\ 17 \end{bmatrix}$$

►

► Given matrices cannot be multiplied

Question No: 19 (Marks: 1) - Please choose one

which of the following is linear equation

► $2x-3y=-6$

► $x+x+x$

► $520x^2y^2$

► $y^2-3=0$

Question No: 20 (Marks: 1) - Please choose one

Given FC = Rs.5000 ,CM= Rs. 30 ,VC= Rs. 150 , Capacity = 320units then BEP in units =

☐☐☐☐☐☐ ► ☐4500 units

☐☐☐☐☐☐ ► ☐167 units

☐☐☐☐☐☐ ► ☐33 units

☐☐☐☐☐☐ ► ☐16 units

Question No: 21 (Marks: 2)

the Discount where price is 2000 and discount rate is 12%.

Find

Solution:

Price = 2000

Discount rate = 12%

Discount = Price * rate

Discount = $2000 \times 12/100$

Discount = 240

Question No: 22 (Marks: 3)

Calculate compound interest earned on Rs. 2250 invested at 8% per annum for 6 years

Solution

compound interest = $(1 - ((1.08)^6)) \times 2250 = 1320$

Question No: 23 (Marks: 5)

If the Basic salary of an employee is Rs 8000 and Allowances are Rs 4,000 . What are the cost on account of 20 casual and 10 sick leaves per year if normal working days per month are 26?

Solution:

Cost of casual leaves per year = $\{18 / (26 \times 12)\} \times 12000 \times 12 = 8307.7$ Rs

Cost of Sick leaves per year = $\{12 / (26 \times 12)\} \times 12000 \times 12 = 5538.5$ Rs.

Question No: 24 (Marks: 10)

The price of medicine inventory is Rs. 500,000.

The series discounts are 20%, 10%, 5%.

What is the single equivalent discount rate?

if Invoice was dated May 1st and 10% discount is offered if invoice is paid up to 10th May. What will be net payment for invoice value of Rs. 500,000 if paid up to 10th May.

Solution

List Price=500,000

$$D1=20\%$$

$$D2=10\%$$

$$D3=5\%$$

$$\text{Net price}=N= L (1-d1)(1-d2)(1-d3)$$

$$= 500,000(1-0.20)(1-0.10)(1-0.05)$$

$$= 342,000$$

Single discounted rate is given by

$$\text{Here we see } N=L(1-d)$$

$$=500000(1-31.6\%)$$

$$=342000$$

So we can say that**31.6%** is a single discounted rate

Payment over 10th of may= 500,000(1- 10%)

= 450,000